

Efficacy of high intensity diode laser as an adjunct to non-surgical periodontal treatment: a randomized controlled trial

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Abstract The high intensity diode laser has been studied in periodontics for the reduction of subgingival bacteria in non-surgical treatment. Our study evaluated the bacterial effect as well as changes in periodontal clinical parameters promoted by root scaling and planing associated with this wavelength. Twenty-seven patients randomly assigned in two groups underwent root scaling and planing on the tested sites, and only the experimental group received the diode laser irradiation. Among the clinical parameters studied, the clinical probing depth (CPD) and the clinical attachment level (CAL) resulted in significant enhancement in the control group when compared with the experimental group ($P=0.014$ and $P=0.039$, respectively). The results were similar for both groups regarding the plaque index

(PI) and bleeding on probing (BP). No significant difference in the microbiological parameters was observed between the control and experimental groups. It was possible to conclude that the high power diode laser adjunct to the non-surgical periodontal treatment did not promote additional effects to the conventional periodontal treatment.

Keywords Chronic periodontitis · Bacterial reduction · High power diode laser · Non-surgical periodontal · Treatment

Introduction

Periodontal diseases result in infections derived from the bacterial biofilm mainly by Gram-negative anaerobic bacteria [1]. The root treatment associated with adequate plaque control plays an important role in periodontal therapy, being a paramount resource and widely applied in this pathologic control [2]. However, in some cases, conventional methods may not be sufficient for periodontal disease control. Under specific circumstances systemic antimicrobials may be used as adjuncts to treatment, leading to significant bacterial reduction and additional clinical benefits to the root scaling and planing procedures [3]. On the other hand, systemic antimicrobials may have potential for adverse reactions and development of resistant bacteria strains [4, 5].

In the past decade studies have demonstrated that laser used as an adjunct to periodontal therapy may promote benefits additional to those of conventional therapy [6–8]. High power diode laser, when compared with those with other wavelengths [9] and tested in several studies regarding its biocompatibility [10], mainly for its capacity to reduce pathological agents [11, 12], has presented characteristics similar to those of the neodymium:yttrium–

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aluminum–garnet (Nd:YAG) laser, besides advantages such as cost and performance. Yilmaz et al. [13] did not find additional benefits when using the high power diode laser together with conventional periodontal treatment. Other studies using the same wavelength led to satisfactory clinical and microbiological results [14, 15], besides promoting bacterial reduction in peri-implantitis sites [16]. This divergence in results might be due to the variation in the different energy parameters. Further studies establishing safe parameters and providing effective results are necessary to verify the real benefit of the laser as a co-adjunct to periodontal treatment.

Thus, the aim of this study was to evaluate the efficacy of the high power diode laser (808 ± 5 nm) as a co-adjunct to root treatment, using more adequate periodontal clinical parameters to achieve bacterial reduction.

Materials and methods

Study design and population

A randomized, split-mouth design, double-blind controlled trial was conducted at the Periodontics Clinic under the approval of the Ethics Committee of the School of Dentistry at the University of São Paulo, Brazil (process no. 116/05).

Based on a 0.8 power setting to detect a significant difference of 1.0 mm in clinical attachment level [$\alpha=0.05$, standard deviation (SD)=1.3 mm, two-sided], 26 volunteers were required for each group. Twenty-eight subjects were screened (10% more) so that we could compensate for any possible loss at follow-up.

The patients were assessed for eligibility prior to the beginning of the study. The following inclusion criteria for the subjects' enrollment in the study were: (1) the presence of chronic periodontitis [17] and (2) the presence of a pair of single-rooted contralateral teeth with a clinical probing depth (CPD) ≥ 5 mm (measured with a University of North Carolina PCP UNC Hu-Friedy periodontal probe, USA) one being the test and the other the control. The exclusion criteria were: (1) teeth that had received, or were undergoing, endodontic treatment; (2) smokers or alcohol and drug users; (3) patients with diseases or systemic conditions that might have acted as modifiers of periodontal disease, such as diabetes, osteoporosis, immunosuppression; (4) patients who had undergone previous periodontal treatment or had received oral antibiotics during the past 6 months.

Preparation and inclusion of patients

Examiner 1 (M.S.), unaware of the intervention on each experimental side, conducted anamnesis, a complete periodontal examination and peri-apical X-rays. All the patients

were given oral hygiene advice and the first survey of clinical and microbiological parameters (baseline) at the appointment, besides previous periodontal treatment at up to four appointments with the exception of the teeth selected for the study, which were treated during the experimental phase.

A 2% lidocaine anesthetic solution was used, and the teeth from the experimental group underwent root scaling and planing with 5/6 and 7/8 Gracey curettes (Hu-Friedy, USA) and were polished with prophylactic paste and a Robinson brush. The calibration for the CPD and the measurement of the distance from the gingival margin to the enamel–cement junction (GM-ECJ) was performed by examiner 1, who was also responsible for the clinical parameter data and for root scaling and planing at the experimental sites, with a 1-week interval between the two examinations. The reproducibility of the examinations was verified through the intraclass correlation coefficient [CPD 0.85, 95% confidence interval (CI) 0.80–0.89; GM-ECJ 0.88, 95% CI 0.84–0.91].

After the periodontal procedure, examiner 2 (V.T.E.A.), responsible for the laser irradiation, tossed a coin to define the patients' assignment in each group.

Diode laser irradiation

A high power diode laser (808 ± 5 nm, Zap Lasers, CA, USA) was used to irradiate the experimental side on day 1 and 1 week after the root planning and scaling procedure. On the control side the laser was applied without activation ('sham' procedure). Prior to the irradiation, Gracey curettes were used for the supra-gingival removal of the biofilm on both sides. Irradiation was accomplished with a 400 μ m fiberoptic delivery system, introduced parallel to the cement surface, with apical–cervical scanning movements for 20 s at 1.5 W in continuous mode (CW). The power density of the fiber tip was 1,193.7 W/cm². The power on the laser display was 1.5 W, but the real output power was measured before each procedure with a power meter. Also, the fibers were cleaved before each irradiation session for the maintenance of its initial physical characteristics. The fiber was introduced by 1 mm less than the value obtained through the probing procedure [6]. For this, an endodontic stop was placed on the fiber to control this irradiation depth. The periodontal pocket was irrigated with saline solution after each session of irradiation. The laser energy was measured before each session with a pyroelectric detector (LM10i) connected to a power/energy meter (Fieldmaster, Coherent, Auburn, USA).

Clinical evaluation

The primary parameter in this study was the clinical attachment level (CAL). The other measures were: CPD,

GM-ECJ, bleeding on probing (BP) and plaque index (PI) [18]. For the probing standardization a stent was made to measure the clinical parameters.

Microbiological evaluation

For subgingival plaque collection the teeth were isolated with cotton rolls and a plaque sample was obtained by the introduction of two sterile no. 40 paper cones inside the pocket for 30 s. Plaque samples were collected at baseline and after 6 weeks. The samples were placed in a vial containing 3 ml of transport medium (VMGA III) [19]. These samples were processed up to 24 h after collection. The vials containing the cones in the VMGA III were incubated at 37°C for 30 min to liquefy the jelly, and then they were immediately homogenized in tube agitators (Fisher Vortex Genie 2, USA).

Aliquots of 100 µl from each sample, some not diluted and some diluted to 1/10 and 1/100 in peptone water, were put into Petri dishes containing a TSBV-selective culture medium (trypticasein soy agar added to horse serum, bacitracin and vancomycin) [20]. After 3 days of incubation at 37°C in an atmosphere containing 5–10% carbon dioxide (CO₂), the colonies were counted and a presumptive identification of *Aggregatibacter actinomycetemcomitans* was made. This identification was based on morphologic aspects of the colony through a stereoscope microscope, morpho-histochemical characteristics under an optical microscope, Gram staining tests and catalase proof.

Aliquots of 100 µl from each sample, diluted to 1/100, 1/1,000 and 1/10,000 in peptone water, were put into Petri dishes containing Brucella agar with 5% goat defibrinated blood, hemin (10 µg/ml, Sigma H-2250) and menadione (1 µg/ml, Sigma M-5625 L) [21]. After 7 days of incubation at 37°C in an anaerobic environment (Anaerobic Jar 2.5 L, Oxoid) the total number of bacterial colony forming units (CFU BT) and black dyed (CFU D) were counted.

The CFU D colonies were presumptively identified based on the lactose fermentation test [22] using the 4-methylumbelliferyl-β-D-galactosidase substrate (MUG; M-1633, Sigma), at a concentration of 1% in dimethyl sulfoxide (D-8799, Sigma). The trypsin activity test [23] was carried out with the fluorogenetic carbobenzoxy L-arginine-7-amino-4-methylcoumarin amine-HCL (CAAM) synthetic compound. The colonies with positive autofluorescence when subjected to a long ultraviolet light wavelength (365 nm Mineralight Lamp UVGL-58, USA) and MUG, and with negative CAAM test results [22] were identified as *Prevotella intermedia*, and the ones with negative results for autofluorescence and the MUG test and positive results for the CAAM test were identified as *Porphyromonas gingivalis* [20].

Statistical analysis

When the samples were normally distributed and the variances were homogeneous, we used Student's *t*-test to verify possible differences between the experimental and control groups. We also used Student's *t*-test for paired samples to compare the average values in each group before and after the laser irradiation. When the samples were not normally distributed and/or the variances were not homogeneous, we used the non-parametric Wilcoxon statistical test. To verify the possibility of an association between the experimental group and the presence of *P. gingivalis*, *P. intermedia* and *A. actinomycetemcomitans* we used the chi square statistical test. Fisher's statistical test was also applied when the chi square test could not be used. A 5% significance level was used for all the statistical tests.

Results

Fifty-two patients were assessed for eligibility. Twenty-four patients were not included, for several reasons (smokers, *n*=11; diabetes, *n*=3; absence of periodontal disease, *n*=10). Twenty-eight patients were included, eight male (29.62%) and 19 female (70.37%), representing an average age of 48.5 years. One patient went through the periodontal treatment but did not return for further appointments and was excluded from the study.

Table 1 shows that there was no significant difference between the groups (*P*=0.10). A significant reduction (*P*<0.001) in the average value of 2.1 mm for the CPD was observed 6 weeks after the treatment. The control group showed a significant reduction (*P*<0.001) in the average value, higher than that of the experimental group (2.4 mm). There was a significant difference (*P*<0.014) between the groups 42 days after the treatment, being positive for the control group.

Regarding the CAL, it was possible to observe that there was no significant difference between the groups (*P*=0.144) at the beginning of the study, but six weeks after the laser irradiation associated with the conventional periodontal treatment a significant reduction (*P*=0.039) in the average value of 1.2 mm for the CAL was noted. A significant reduction (*P*<0.001) in the average value of 1.9 mm for the CAL was observed in the control group, higher than that in the experimental group (*P*=0.039).

No significant difference between the groups was observed at the beginning of the study in the GM-ECJ (*P*=0.47). A significant increase in the average value of the gingival retraction of 0.6 mm (*P*=0.019) was observed in the experimental group and in the control group; this significant increase in average value was 0.5 mm (*P*=0.009). However, no significant difference between the

Table 1 Means and standard deviations of the clinical probing depth (CPD), distance from the gingival margin to the enamel–cement junction (GM-ECJ), clinical attachment level (CAL), bleeding on probing (BP) and plaque index (PI) for the test (*T*) group and control (*C*). *SS* statistical significance

Parameter	Mean ± SD at baseline	Mean ± SD after 6weeks	<i>P</i> ^a
CPD			
T	6.2±1.4	4.1±1.6	<0.001 ^c
C	5.8±1.0	3.4±1.4	<0.001 ^c
<i>P</i> ^b	0.1	0.014 ^c	
GM-ECJ			
T	0.8±1.2	1.4±1.9	<0.019 ^c
C	0.6±1.3	1.1±1.2	<0.009 ^c
<i>P</i> ^b	0.47	0.43	
CAL			
T	6.9±1.9	5.7±2.6	<0.001 ^c
C	6.4±1.5	4.5±1.8	<0.001 ^c
<i>P</i> ^b	0.144	0.039 ^c	
SS (%)			
T	100.0±0.0	51.8±50.9	<0.001 ^c
C	96.2±19.5	40.7±50.0	<0.001 ^c
<i>P</i> ^b	0.32	0.42	
PI			
T	1.6±0.9	0.2±0.3	<0.001 ^c
C	1.8±0.7	0.2±0.3	<0.001 ^c
<i>P</i> ^b	0.08	1.00	

^a Between baseline and 6 weeks after

^b Between test and control groups

^c Statistical significance at 5%

average values in both groups 6 weeks after the treatment was observed ($P=0.443$).

There was a significant PI and BP reduction, both in the experimental group ($P<0.001$) and control group ($P<0.001$). However, no significant difference was observed between the experimental groups regarding these variables at any time in the study.

No significant difference between the two groups at the beginning of the study ($P=0.50$) regarding the number of CFU BT was observed (Table 2). A significant reduction in the numbers of CFU BT in the experimental group ($P<0.001$) and control group ($P<0.001$) was noted, but there was no significant difference between the groups 6 months after the treatment ($P=0.55$).

One can also note that there was no significant difference between the two groups at the beginning ($P=0.06$) regarding the total number of CFU D (Table 2). There was a significant reduction in this number from the first analysis to the second in the experimental group ($P=0.014$), this reduction being higher in the experimental group, but no statistically significant difference between the groups was found ($P=0.30$).

For the numbers of *A. actinomycetemcomitans* CFUs, there was no significant difference between the groups at the beginning ($P=0.89$) (Table 3). A reduction was noted in both groups, being higher in the control group, although this difference was not statistically significant ($P=0.10$).

There was no association regarding the group and the presence of *P. gingivalis* and *P. intermedia* at any time,

showing that the presence of bacteria was not significantly higher in either of the experimental groups.

Discussion

The literature shows that lasers have several applications in dentistry. Especially in periodontics, the studies show that their bactericidal effect is the main reason for their use as adjuncts to conventional periodontal treatment and/or their use in special systemic conditions [8, 24]. The results of our study demonstrated that high power diode laser associated with root scaling and planing did not promote additional enhancement of the clinical parameters. Both groups presented positive results after the treatment, but the control group showed better results regarding CPD and CAL. There was no difference for the other parameters evaluated (GM-ECJ, BP, and PI). These results are not in agreement with those obtained by Borrajo et al. [15], who did not demonstrate differences between the groups for CPD, or those of Moritz et al. [14] or Kamma et al. [24] who demonstrated superior results in the groups where the laser was used with conventional periodontal treatment.

The results from Kreisler et al. [25] regarding the effects of laser treatment on the survival rate of human gingival fibroblasts may help to justify our findings. It states that the indications for the use of a diode laser for decontamination of the periodontal pockets may cause damage to the surrounding soft tissue if the power and treatment duration

Table 2 Means and standard deviations in the test group and control regarding the numbers of colony forming units (CFUs) and the CFUs of dark-pigmented bacteria at different times

Number of CFUs of all bacteria	Mean $\pm$ SD at baseline	Mean $\pm$ SD after 6weeks	P^a
Test	$24.9 \times 10^4 \pm 25.7 \times 10^4$	$3.6 \times 10^4 \pm 7.3 \times 10^4$	$<0.0001^c$
Control	$28.8 \times 10^4 \pm 40.5 \times 10^4$	$4.6 \times 10^4 \pm 4.7 \times 10^4$	0.005^c
P^b	0.50	0.55	
CFU of dark pigmented bacteria	Mean $\pm$ SD at baseline	Mean $\pm$ SD after 6 weeks	P^a
Test	$12.0 \times 10^3 \pm 28.5 \times 10^3$	$1.5 \times 10^3 \pm 5.1 \times 10^3$	0.003^c
Control	$43.9 \times 10^3 \pm 101.6 \times 10^3$	$5.3 \times 10^3 \pm 15.8 \times 10^3$	0.014^c
P^b	0.06	0.30	

^a Between baseline and 6 weeks after^b Between test and control groups^c Statistical significance at 5%

are not adequate. Kreisler et al. [10] also affirmed that diode laser irradiation at 1 W for 20 s did not have a substantially positive effect on the new attachment of periodontal ligament cells. Yilmaz et al. [13] also did not demonstrate an additional benefit of the diode laser when compared with conventional treatment.

The results of our study showed that high power diode laser associated with conventional periodontal treatment and isolated root scaling and planing had promoted in a similar way a reduction in the average numbers of all CFU D and *A. actinomycetemcomitans* bacteria 6 weeks after the treatment. These findings are opposed to those of Moritz et al. [11], [14] and Poleník [26], who observed additional bacterial reduction after high power diode laser irradiation associated with non-surgical periodontal therapy.

No association between the experimental group and the presence of the bacteria analyzed in the study was observed, meaning that the presence of bacteria was not significantly greater in either of the experimental groups.

Chan and Chien [27] confirmed that an in vivo Nd:YAG laser bactericidal effect was due to an interaction of this wavelength with pigments [28]; thus, the black pigmented pathogenic periodontal bacteria would absorb this energy because this dark pigment would act as a laser-absorbing chromophore.

It is known that *P. gingivalis* and *P. intermedia* produce black pigments in agar–blood Brucella media, but there is no in vivo research in the literature that demonstrates the production of these pigments. It is possible to affirm that the hemoglobin in the soft tissue of periodontal pockets is an absorber chromophore for the high power diode laser and acts as an endogenous dye, enhancing the laser's effect at this site.

The literature does not show any standard procedures regarding energy, group distribution, modes of irradiation and time taken for bacterial reduction in the periodontal pocket, which hinders the precise analysis of the results. The power settings tested and used varied from 1 W in a continuous mode [25] to 2.5 W in a pulsed mode [14]. Bach et al. [29] and Moritz et al. [14] studied the time variation as a function of clinical probing depth.

The clinical parameters in this study (1.5 W, CW, 20 s) and the references for a safe fiberoptic angle position inside the periodontal pocket during the laser irradiation followed those of the in vitro study by Kreisler et al. [30], who used a high power diode laser to evaluate and quantify morphological changes in the root cement.

The use of a power meter is paramount when one is working with lasers, in order to verify the real amount of energy at the tip of the fiberoptic delivery system, because the parameter values selected on the display of the laser

Table 3 Means and standard deviations of the test group and control regarding the numbers of colony forming units (CFUs) of *A. actinomycetemcomitans* (A.a) at different times

CFU A.a	Mean $\pm$ SD at baseline	Mean $\pm$ SD after 6weeks	P^a
Test	87.0 ± 355.6	82.4 ± 300.3	0.71
Control	208.2 ± 939.5	11.1 ± 57.7	0.28
P^b	0.89	0.10	

^a Between baseline and 6 weeks after^b Between test and control groups

device lose energy as it is transmitted through a fiberoptic (20% transmission factor).

The search for answers regarding the effects of lasers when applied to periodontal tissues, where different structures interact harmonically, is a challenge for dentistry. Further studies are necessary to establish adequate conditions for irradiation, such as the use of a water refrigeration system for the decontamination of periodontal pockets.

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